

C EXP4.c > enqueue()

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1  #include <stdio.h>
2  #define MAX 6
3  int queue[MAX];
4  int front = -1;
5  int rear = -1;
6  void enqueue();
7  void dequeue();
8  void display();
9  int main()
10 {
11
12     int choice = 1;
13     while (choice < 4 && choice != 0) {
14         printf("\nPress 1: Insert an element");
15         printf("\nPress 2: Delete an element");
16         printf("\nPress 3: Display the elements");
17         printf("\nPress 0: Exit");
18         printf("\nEnter your choice: ");
19         scanf("%d", &choice);
20         switch (choice)
21         {
22             case 1:
23                 enqueue();
24                 break;
25             case 2:
26                 dequeue();
27                 break;
28             case 3:
29                 display();
30                 break;
31             case 0:
32                 printf("Exiting...\n");
33                 break;
34             default:
35                 printf("Invalid choice! Try again.\n");
36         }
37     }
```

```

40 void enqueue() {
41     int element;
42     if ((rear + 1) % MAX == front){
43         printf("Queue is overflow..\n");
44         return;
45     }
46     printf("Enter the element to be inserted: "); scanf("%d", &element);
47     if (front == -1 && rear == -1){
48         front = 0;
49         rear = 0;
50     }
51     else {
52         rear = (rear + 1) % MAX;
53     }
54     queue[rear] = element;
55     printf("Inserted -> %d\n", element);
56 }
57 void dequeue() {
58     if (front == -1) {
59         printf("Queue is underflow..\n");
60         return;
61     }
62     printf("The dequeued element is %d\n", queue[front]);
63     if (front == rear) {
64         front = -1;
65         rear = -1;
66     }
67     else {
68         front = (front + 1) % MAX;
69     }
70 }
71 void display(){
72     if (front == -1){
73         printf("Queue is empty..\n"); return;
74     }
75     printf("Elements in the queue are: "); int i = front;

```

```

void display(){
    if (front == -1){
        printf("Queue is empty..\n"); return;
    }
    printf("Elements in the queue are: "); int i = front;
    while (1) {
        printf("%d ", queue[i]);
        if (i == rear) {
            break;
        }
        i = (i + 1) % MAX;
    }
    printf("\n");
}

```

Press 1: Insert an element
Press 2: Delete an element
Press 3: Display the elements
Press 0: Exit
Enter your choice: 1
Enter the element to be inserted: 23
Inserted -> 23

Press 1: Insert an element
Press 2: Delete an element
Press 3: Display the elements
Press 0: Exit
Enter your choice: 3
Elements in the queue are: 23

Press 1: Insert an element
Press 2: Delete an element
Press 3: Display the elements
Press 0: Exit
Enter your choice: 2
The dequeued element is 23

Press 1: Insert an element
Press 2: Delete an element
Press 3: Display the elements
Press 0: Exit
Enter your choice: 3
Queue is empty..

Press 1: Insert an element
Press 2: Delete an element
Press 3: Display the elements
Press 0: Exit
Enter your choice:
0
Exiting...